

# [ShadowFox Data Science Internship]

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[Intermediate Level Task]

[Air Quality Index (AQI) Analysis Using Python]

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[SHADOWFOX DATA SCIENCE INTERNSHIP]

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## Introduction:

Air Quality Index (AQI) is a numerical scale used to measure and communicate the quality of air in a specific location. It indicates the level of air pollution and its potential and its potential impact on human health. The AQI is calculated based on the concentration of major air pollutants such as PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, SO<sub>2</sub>, CO, and O<sub>2</sub>.

## Objectives:

The main objectives of this project are:

- . To analyse the Air Quality Index (AQI) dataset.
- . To clean and preprocess the dataset.
- . To understand the relationship between AQI and different air pollutants.
- . To visualize air quality trends using graphs and charts.
- . To identify patterns, correlations, and possible outliers in the data.

To draw meaningful conclusions that support environmental awareness and decision-making.

## Dataset Description:

| Column Name       | Description                 |
|-------------------|-----------------------------|
| 1.Date            | 1. Date of observation      |
| 2.Month           | 2. Month of observation     |
| 3.Year            | 3. Year of observation      |
| 4.Holidays_count  | 4. Number of holidays       |
| 5.Days            | 5. Day of the week          |
| 6.PM2.5           | 6.Fine particulate matter   |
| 7.PM10            | 7. Particulate matter       |
| 8.NO <sub>2</sub> | 8. Nitrogen Dioxide         |
| 9.SO <sub>2</sub> | 9. Sulfur Dioxide           |
| 10. CO            | 10. Carbon Monoxide         |
| 11. Ozone         | 11. Ozone Concentration     |
| 12. AQI           | 12. Air Quality Index value |

The dataset contains 1461 observations and 12 variables, making it suitable for exploratory data analysis(EDA), visualization, and statistical interpretation.

## Tools and Technologies Used:

- . Python
- . Jupyter Notebook
- . Pandas
- . NumPy
- . Matplotlib
- . Seaborn

## Data Loading and Exploratory Data Analysis(EDA):

### 1.Import Required Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

### 2.Uploaded CSV File Dataset

**final\_dataset.csv**(text/csv) - 78368 bytes, last  
modified: 7/21/2026 - 100% done

Saving final\_dataset.csv to final\_dataset.csv

### 3. Import Dataset

```
import pandas as pd
```

```
df = pd.read_csv("final_dataset.csv")
```

```
df.head()
```

|   | Date | Month | Year | Holidays_Count | Days | PM2.5  | PM10   | NO2    | SO2   | CO   | Ozone | AQI |
|---|------|-------|------|----------------|------|--------|--------|--------|-------|------|-------|-----|
| 0 | 1    | 1     | 2021 | 0              | 5    | 408.80 | 442.42 | 160.61 | 12.95 | 2.77 | 43.19 | 462 |
| 1 | 2    | 1     | 2021 | 0              | 6    | 404.04 | 561.95 | 52.85  | 5.18  | 2.60 | 16.43 | 482 |
| 2 | 3    | 1     | 2021 | 1              | 7    | 225.07 | 239.04 | 170.95 | 10.93 | 1.40 | 44.29 | 263 |
| 3 | 4    | 1     | 2021 | 0              | 1    | 89.55  | 132.08 | 153.98 | 10.42 | 1.01 | 49.19 | 207 |
| 4 | 5    | 1     | 2021 | 0              | 2    | 54.06  | 55.54  | 122.66 | 9.70  | 0.64 | 48.88 | 149 |

### 4. Load the Dataset

```
df = pd.read_csv("final_dataset.csv")
```

### 5. Display the First Five Rows

```
df.head()
```

|   | Date | Month | Year | Holidays_Count | Days | PM2.5  | PM10   | NO2    | SO2   | CO   | Ozone | AQI |
|---|------|-------|------|----------------|------|--------|--------|--------|-------|------|-------|-----|
| 0 | 1    | 1     | 2021 | 0              | 5    | 408.80 | 442.42 | 160.61 | 12.95 | 2.77 | 43.19 | 462 |
| 1 | 2    | 1     | 2021 | 0              | 6    | 404.04 | 561.95 | 52.85  | 5.18  | 2.60 | 16.43 | 482 |
| 2 | 3    | 1     | 2021 | 1              | 7    | 225.07 | 239.04 | 170.95 | 10.93 | 1.40 | 44.29 | 263 |
| 3 | 4    | 1     | 2021 | 0              | 1    | 89.55  | 132.08 | 153.98 | 10.42 | 1.01 | 49.19 | 207 |
| 4 | 5    | 1     | 2021 | 0              | 2    | 54.06  | 55.54  | 122.66 | 9.70  | 0.64 | 48.88 | 149 |

## 6. Display the Last Five Rows

df.tail()

|      | Date | Month | Year | Holidays_Count | Days | PM2.5 | PM10   | NO2   | SO2   | CO   | Ozone | AQI |
|------|------|-------|------|----------------|------|-------|--------|-------|-------|------|-------|-----|
| 1456 | 27   | 12    | 2024 | 0              | 5    | 58.43 | 249.17 | 41.69 | 65.89 | 0.99 | 36.25 | 263 |
| 1457 | 28   | 12    | 2024 | 0              | 6    | 33.83 | 150.77 | 33.31 | 66.14 | 0.79 | 35.19 | 113 |
| 1458 | 29   | 12    | 2024 | 1              | 7    | 31.21 | 139.75 | 27.01 | 65.94 | 0.57 | 35.88 | 142 |
| 1459 | 30   | 12    | 2024 | 0              | 1    | 38.01 | 152.83 | 29.12 | 65.16 | 0.55 | 38.38 | 116 |
| 1460 | 31   | 12    | 2024 | 0              | 2    | 80.42 | 318.96 | 40.37 | 64.98 | 0.84 | 39.93 | 209 |

## 7. Check Dataset Shape:

df.shape

(1461, 12)

The function returns the number of rows and columns present in the dataset.

## Display Column Names:

df.columns

Index(['Date', 'Month', 'Year', 'Holidays\_Count', 'Days',  
'PM2.5', 'PM10',

'NO2', 'SO2', 'CO', 'Ozone', 'AQI'],

dtype='object')

## 4. Dataset Information:

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1461 entries, 0 to 1460
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Date                  1461 non-null  int64  
 1   Month                 1461 non-null  int64  
 2   Year                  1461 non-null  int64  
 3   Holidays_Count        1461 non-null  int64  
 4   Days                  1461 non-null  int64  
 5   PM2.5                 1461 non-null  float64 
 6   PM10                  1461 non-null  float64 
 7   NO2                   1461 non-null  float64 
 8   SO2                   1461 non-null  float64 
 9   CO                    1461 non-null  float64 
10   Ozone                 1461 non-null  float64 
11   AQI                   1461 non-null  int64  
dtypes: float64(6), int64(6)
memory usage: 137.1 KB
```



5. Statistical Summary:

```
df.describe()
```

|       | Date        | Month       | Year        | Holidays_Count | Days        | PM2.5       | PM10        | NO2         | SO2         | CO          | Ozone       | AQI         |
|-------|-------------|-------------|-------------|----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| count | 1461.000000 | 1461.000000 | 1461.000000 | 1461.000000    | 1461.000000 | 1461.000000 | 1461.000000 | 1461.000000 | 1461.000000 | 1461.000000 | 1461.000000 | 1461.000000 |
| mean  | 15.729637   | 6.522930    | 2022.501027 | 0.189596       | 4.000684    | 90.774538   | 218.219261  | 37.184921   | 20.104921   | 1.025832    | 36.338871   | 202.210815  |
| std   | 8.803105    | 3.449884    | 1.118723    | 0.392116       | 2.001883    | 71.650579   | 129.297734  | 35.225327   | 16.543659   | 0.608305    | 18.951204   | 107.801076  |
| min   | 1.000000    | 1.000000    | 2021.000000 | 0.000000       | 1.000000    | 0.050000    | 9.690000    | 2.160000    | 1.210000    | 0.270000    | 2.700000    | 19.000000   |
| 25%   | 8.000000    | 4.000000    | 2022.000000 | 0.000000       | 2.000000    | 41.280000   | 115.110000  | 17.280000   | 7.710000    | 0.610000    | 24.100000   | 108.000000  |
| 50%   | 16.000000   | 7.000000    | 2023.000000 | 0.000000       | 4.000000    | 72.060000   | 199.800000  | 30.490000   | 15.430000   | 0.850000    | 32.470000   | 189.000000  |
| 75%   | 23.000000   | 10.000000   | 2024.000000 | 0.000000       | 6.000000    | 118.500000  | 297.750000  | 45.010000   | 26.620000   | 1.240000    | 45.730000   | 284.000000  |
| max   | 31.000000   | 12.000000   | 2024.000000 | 1.000000       | 7.000000    | 1000.000000 | 1000.000000 | 433.980000  | 113.400000  | 4.700000    | 115.870000  | 500.000000  |

6. Check Missing Values:

```
df.isnull().sum()
```

|                |   |
|----------------|---|
|                | 0 |
| Date           | 0 |
| Month          | 0 |
| Year           | 0 |
| Holidays_Count | 0 |
| Days           | 0 |
| PM2.5          | 0 |
| PM10           | 0 |

|       |   |
|-------|---|
|       | 0 |
| NO2   | 0 |
| SO2   | 0 |
| CO    | 0 |
| Ozone | 0 |
| AQI   | 0 |

**dtype:** int64

## 7. Remove Duplicate Rows (If Any)

```
df.drop_duplicates(inplace=True)
```

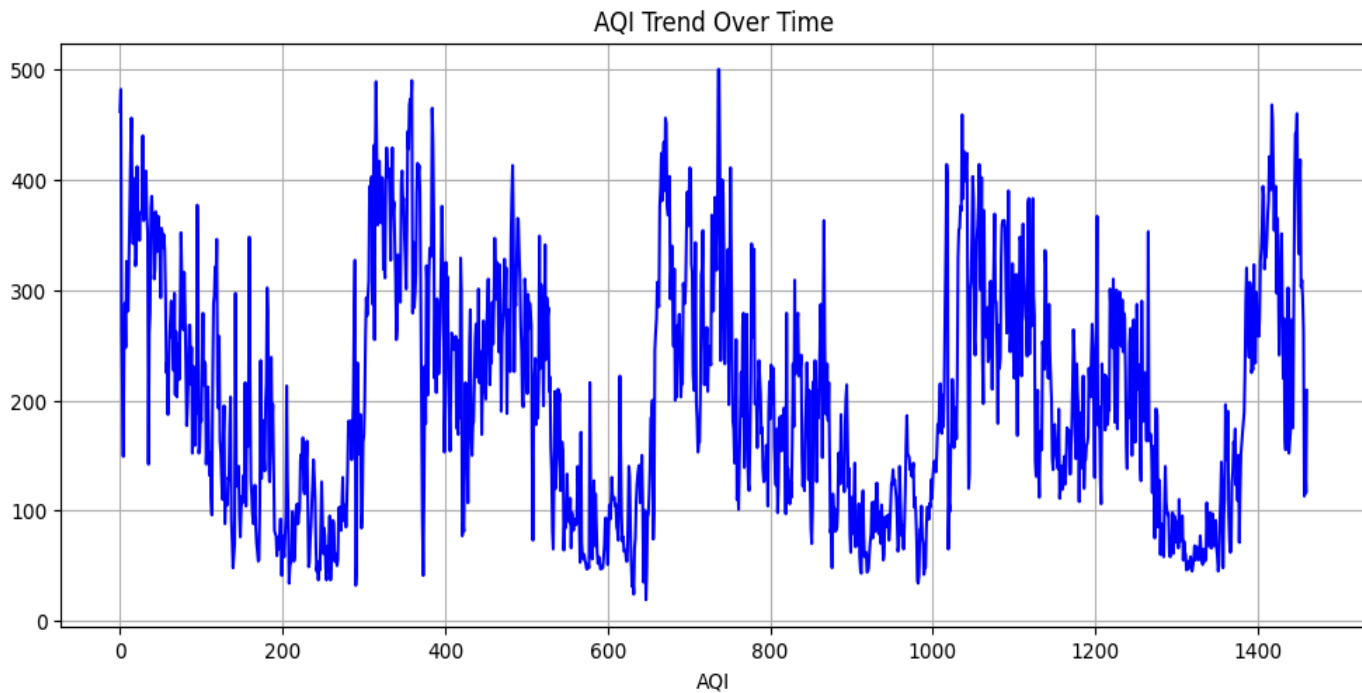
## 8. Data Cleaning

```
df.dropna(inplace=True)
```

# Data Visualization:

## 1. Line Plot

```
import matplotlib.pyplot as plt
plt.figure(figsize=(12,5))
plt.plot(df["AQI"], color="blue")
plt.title("AQI Trend Over Time")
plt.xlabel("AQI")
plt.grid(True)
plt.show()
```

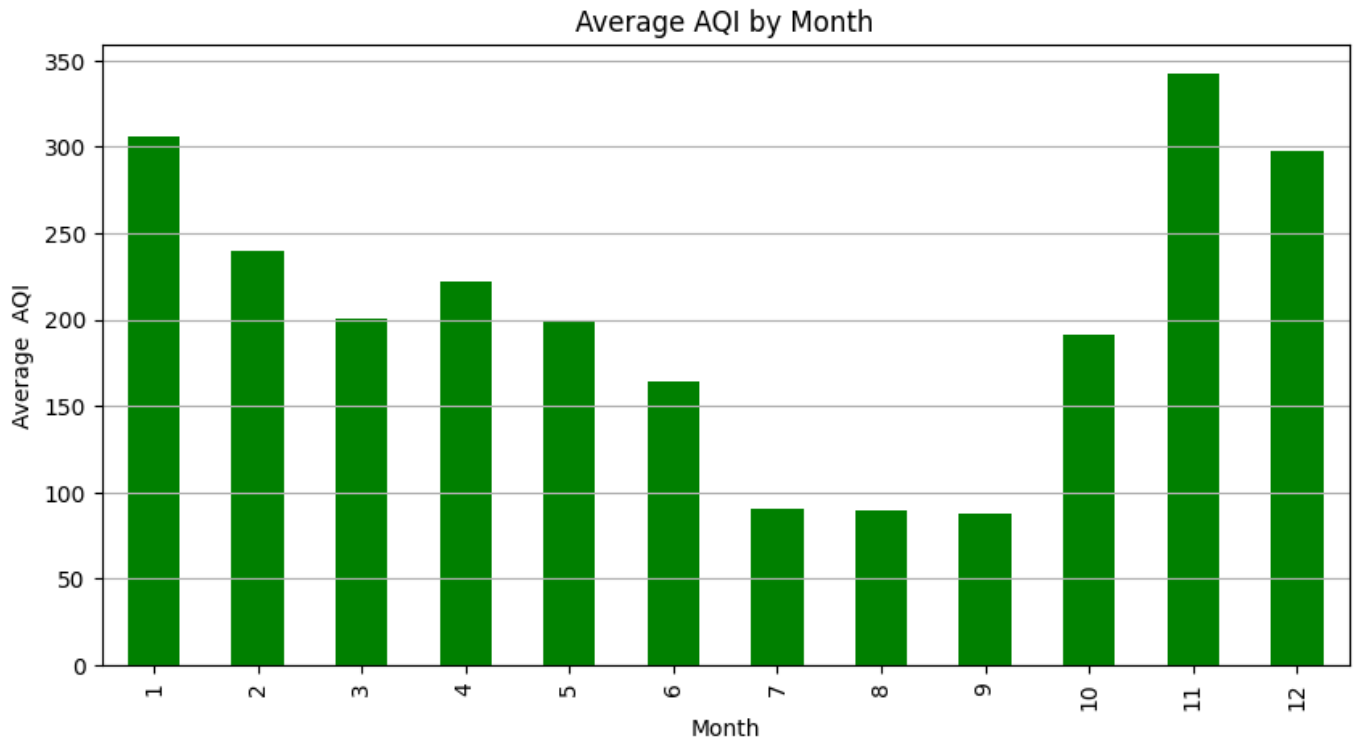


## Explanation

A line plot is used to visualize how AQI changes over time. It helps identify increasing or decreasing pollution trends and seasonal variations.

## 2. Bar Chart

```
plt.figure(figsize=(10, 5))
df.groupby("Month")["AQI"].mean().plot(
    kind="bar", color="green")
plt.title("Average AQI by Month")
plt.xlabel("Month")
plt.ylabel("Average AQI")
plt.grid(axis="y")
plt.show()
```

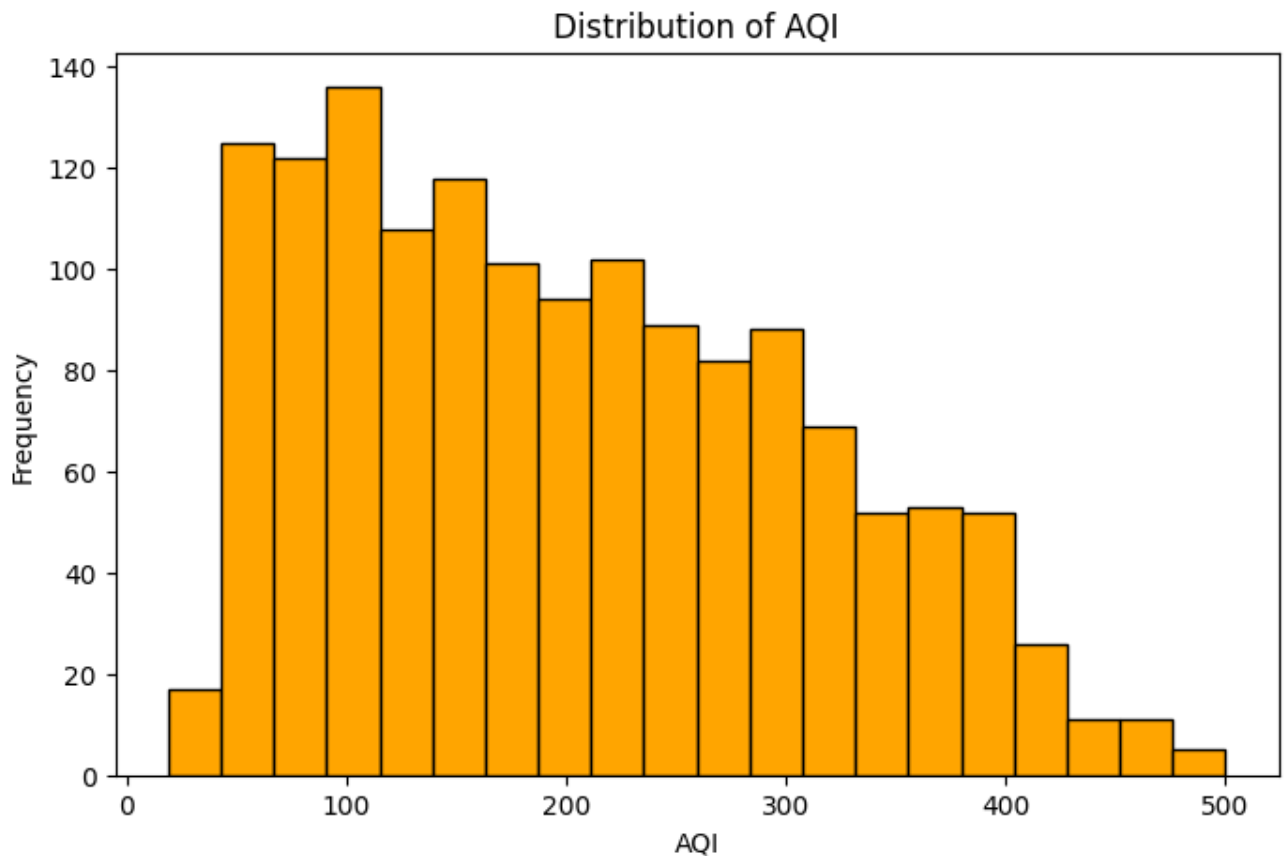


## Explanation

This graph compares the average AQI of each month and helps identify the months with the highest and lowest pollution levels.

### 3. Histogram

```
plt.figure(figsize=(8,5))
plt.hist(df["AQI"], bins=20,
color="orange", edgecolor="black")
plt.title("Distribution of AQI")
plt.xlabel("AQI")
plt.ylabel("Frequency")
plt.show()
```

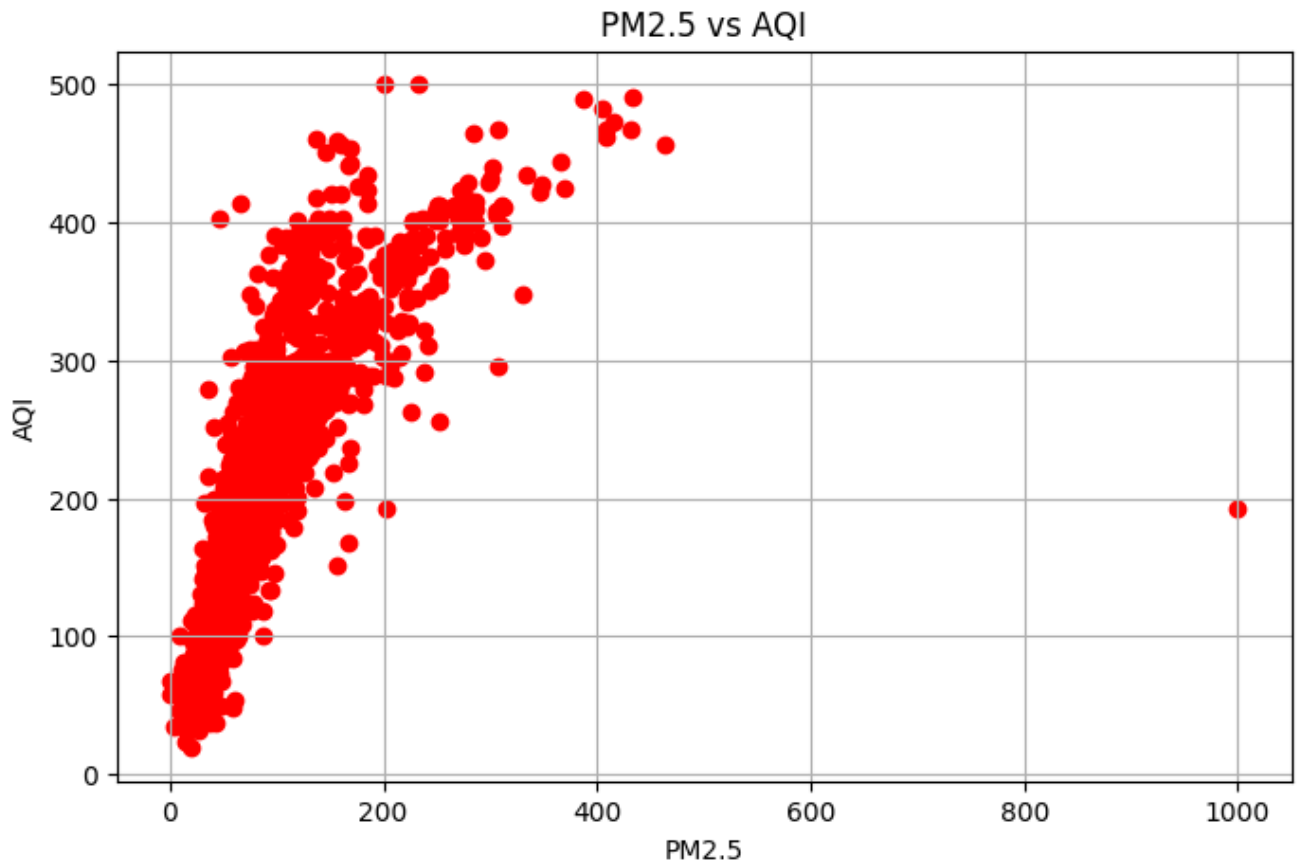


## Explanation

The histogram shows how AQI values are distributed across the dataset.

## 4. Scatter Plot

```
plt.figure(figsize=(8,5))
plt.scatter(df["PM2.5"], df["AQI"],
            color="red")
plt.title("PM2.5 vs AQI")
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.grid(True)
plt.show()
```



## Explanation

This graph helps determine whether higher PM2.5 levels are associated with higher AQI values.

## Project Insights:

After analyzing the AQI dataset, the following insights were observed:

1. Air Quality Index (AQI) values vary significantly across different dates, indicating fluctuations in air pollution levels.

2. PM2.5 and PM10 show a strong positive relationship with AQI, suggesting that particulate matter has a major impact on air quality.
3. Higher concentrations of pollutants generally result in higher AQI values.
4. The heatmap reveals correlations among different pollutants, helping identify which pollutants influence AQI the most.
5. Some months have consistently higher average AQI values, indicating seasonal variations in air pollution.

## Discussion:

The analysis demonstrates how different air pollutants affect the Air Quality Index. Among all pollutants, PM2.5 and PM10 contribute significantly to poor air quality. Seasonal changes and environmental conditions also influence AQI levels.

## **Conclusion:**

This project successfully analyzed the Air Quality Index (AQI) dataset using Python. Data preprocessing, exploratory data analysis, and visualization techniques were applied to understand pollution trends and relationships among different pollutants.

The visualizations showed that particulate matter (PM2.5 and PM10) has a significant impact on AQI values. Correlation analysis and graphical representations helped identify important patterns, seasonal changes, and data distribution.

## **Future Scope:**

- . Build a Machine Learning model to predict AQI.
- . Create an interactive dashboard using Streamlit or Power BI.
- . Analyze air quality data from multiple cities.
- . Perform real-time AQI monitoring using live APIs.
- . Develop an early warning system for poor air quality.



## **References:**

- . Pandas Documentation – <https://pandas.pydata.org/>
- . Matplotlib Documentation – <https://matplotlib.org/>
- . Seaborn Documentation – <https://seaborn.pydata.org/>
- . Python Documentation – <https://docs.python.org/3/>
- . Air Quality Index (AQI) Guidelines – Central Pollution Control Board (CPCB), India.